Tactix: Tier-D Protocol Preparation and Open Gates

Report to Andrew

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Outcome

The future 1,166-instance Tier-D Tactix comparison now has a preparation path for automatic certificates, complete frozen checker pins, anonymous node-local scratch and the requested first-winner per-task limits. This is a tooling checkpoint, *not* a submitted or accepted Tier-D run. H2/N2, the cumulative coverage and performance gates, N1/N3/N4, and the online-learning refactors remain open. The detailed change and dry-run record is AM433.

What changed and what was verified

The older dedicated Tier-D launcher still forces `-drat`; that is a compatibility mode, not the solver's default automatic proof behavior. The anonymous-scratch preparer now selects explicit `-dsr` or `-auto-proof`, pins the complete checker bundle against frozen hashes, and budgets the full solve plus checker window. Automatic output is content-routed by the existing runtime: startup PBP to VeriPB/CakePB, witnessed DSR to no-option `dsr-trim`/Trestle, and witness-free DSR to DRAT/CakeLPR, with original-CNF checking.

An unsubmitted full-union dry run generated 1,166 tasks, an unthrottled `0-1165` array on `cpu-dense-hwulab-q`, four CPUs, 32 GiB, 5,000 solver seconds, 45,000 checker seconds, and a 14:15:00 scheduler allowance. Its checker bundle passed the runtime pin contract. The first dry run exposed a preparer-local source-path shadowing bug; after repair, a second dry run verified the exact source-tree path and resource fields. Both temporary campaigns were never submitted and were moved to recoverable trash. Focused protocol tests, 17 runtime/scratch unit tests, Python compilation and the memory-enforcement test pass. Compute-node validation of a final accepted binary is still required before submission.

Live evidence and next actions

At 19:34 UTC the matched AM427 Sweep Tier-C comparison had 58/60 result records, 29 complete pairs, and no recorded proof/model failure or paired verdict/proof-size mismatch. The final `aes_decry_2_rounds` pair was still checking; aggregate job `1723300` was dependent. The current-source full Debug job `1723391` remained queued. The 29-pair solver total favored the candidate by 48.311 seconds, driven mainly by one case, and cannot establish N4 or a stable performance gain.

The canonical no-option checker still rejects the separate H2 proof because it retains implied-unit reason clauses that Tactix has deleted. Any repair must handle the checker-visible formula generally; an alternative checker option needs an explicit contract decision and broad positive/negative audit. Continue the frozen Tier-C/Debug checks and affected fuzz, resolve H2, then complete the retained Rank/B+C/WV and N1–N4 gates before launching Tier D. The prior AM430 learning review stands: production online learning needs shared slot-control, owned switchable sub-policies, and live default policy session ownership.